AI/ML intern DAY 9

Research on Fashion Datasets and Data Preparation for Virtual Try-On Systems

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Date: 21-05-2026

Introduction

Artificial Intelligence applications in the fashion industry rely heavily on large-scale, well-organized datasets. Virtual Try-On systems, fashion recommendation engines, garment recognition models, and clothing segmentation networks require thousands of labelled images to achieve high accuracy.

The quality of an AI model is directly influenced by the quality of the data used for training. Therefore, collecting, cleaning, organizing, and annotating fashion datasets are critical steps in the development of intelligent fashion applications.

This research focuses on understanding the importance of fashion datasets, studying popular public datasets used in Virtual Try-On systems, and exploring the processes involved in dataset preparation.

Objectives

The objectives of this research are:

- Study the importance of datasets in AI.
- Analyse popular fashion datasets.
- Understand dataset collection methods.
- Explore data cleaning techniques.
- Study metadata generation and annotation processes.
- Learn how testing samples are prepared.

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1. Importance of Fashion Datasets

Artificial Intelligence models require large amounts of training data to recognize clothing patterns and understand garment characteristics.

Fashion datasets help AI systems perform tasks such as:

- Clothing Classification

- Garment Segmentation
- Virtual Try-On
- Fashion Recommendation
- Landmark Detection
- Human Parsing

A properly organized dataset improves both model accuracy and training efficiency.

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2. DeepFashion2 Dataset

Overview

DeepFashion2 is one of the largest and most comprehensive fashion datasets available for computer vision research.

It contains a wide variety of clothing images collected under different conditions.

Features

- Large image collection
- Multiple clothing categories
- Landmark annotations
- Bounding boxes
- Segmentation masks

Applications

- Virtual Try-On Systems
- Clothing Detection
- Fashion Retrieval
- Product Recommendation

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3. VITON-HD Dataset

Overview

VITON-HD is an advanced dataset specifically designed for high-resolution Virtual Try-On research.

It provides paired images of people and garments that help train garment transfer models.

Characteristics

- High-resolution images

- Realistic body poses
- Clothing alignment information
- Garment masks

Applications

- HR-VITON
- StableVITON
- AI Fitting Systems

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4. Dataset Collection Process

The dataset collection stage involves gathering images from reliable sources.

Typical sources include:

- Fashion websites
- Public AI repositories
- Research datasets
- E-commerce platforms

During collection, diversity is important.

Images should contain:

- Different body types
- Various clothing styles
- Multiple poses
- Different lighting conditions

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5. Data Cleaning

Raw datasets often contain errors and inconsistencies.

Data cleaning improves overall dataset quality.

Common Cleaning Operations

Duplicate Removal

Duplicate images are identified and removed.

Corrupted File Detection

Damaged or unreadable images are excluded.

Image Quality Verification

Low-resolution or blurry images are filtered.

Category Validation

Images are checked to ensure they belong to the correct clothing category.

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6. Metadata Generation

Overview

Metadata refers to structured information that describes the properties and characteristics of each image in a dataset. Instead of analysing the image itself, AI models and dataset management systems use metadata to organize, search, and process fashion data efficiently.

Metadata acts as a digital description card for every clothing image and helps researchers quickly identify important information without manually opening each file.

Importance of Metadata

Metadata provides several advantages:

- Simplifies dataset organization.
- Improves data retrieval speed.
- Supports AI model training.
- Enables efficient filtering and categorization.
- Helps maintain dataset consistency.

For large-scale fashion datasets containing thousands of images, metadata becomes an essential component of data management.

Common Metadata Fields

Metadata Field	Description
Image ID	Unique image identifier
File Name	Original image file name
Clothing Category	Shirt, Pants, Dress, Jacket, etc.
Gender	Male, Female, Unisex
Pose Type	Front, Side, Back
Image Resolution	Width × Height

Background Type	Plain or Complex
Dataset Split	Training, Validation, Testing
File Format	JPG, PNG

Role in Virtual Try-On Systems

Metadata allows AI models to quickly retrieve images with similar characteristics during training and evaluation.

For example, a Virtual Try-On model can easily filter all "Front View Male T-Shirts" without scanning every image manually.

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7. Annotation Process

Overview

Annotation is the process of manually or automatically labelling different parts of an image so that Artificial Intelligence models can understand visual information.

Annotations serve as the ground truth for supervised machine learning algorithms.

Importance of Annotation

High-quality annotations improve:

- Model accuracy
- Clothing detection
- Landmark localization
- Segmentation quality
- Garment alignment

Poor annotations often lead to incorrect model predictions.

Types of Annotations Used in Fashion Datasets

1. Bounding Box Annotation

Bounding boxes define the approximate location of a garment inside an image.

Example:

- Shirt
- Pants

- Dress

Each clothing item is enclosed within a rectangular boundary.

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2. Landmark Annotation

Landmarks identify important points on a garment.

Typical landmarks include:

- Collar
- Left Sleeve
- Right Sleeve
- Waistline
- Hemline

These landmarks are extremely important for Virtual Try-On applications because they help align garments with the human body.

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3. Segmentation Mask Annotation

Segmentation masks assign labels to every pixel.

Example:

- Background
- Upper Clothing
- Lower Clothing
- Arms
- Face

This enables precise garment extraction and body understanding.

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4. Attribute Annotation

Attribute labels describe clothing characteristics such as:

- Colour
- Pattern
- Sleeve Length
- Neck Type
- Fabric Style

These attributes are widely used in fashion recommendation systems.

Annotation Workflow

The general annotation process consists of:

Image Collection
↓
Image Verification
↓
Bounding Box Labelling
↓
Landmark Annotation
↓
Segmentation Mask Creation
↓
Attribute Labelling
↓
Quality Checking
↓
Final Dataset Preparation

Annotation Challenges

Several challenges arise during dataset annotation:

- Complex backgrounds.
- Overlapping garments.
- Multiple people in one image.
- Inconsistent lighting conditions.
- Human annotation errors.

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8. Testing Sample Preparation

Testing samples are separated from training data.

The testing dataset should contain:

- Unseen images
- Multiple clothing categories
- Different body poses

- Various backgrounds

Proper testing ensures fair evaluation of AI models.

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9. Dataset Organization

A well-structured dataset improves workflow efficiency.

Example structure:

Dataset

- Train
    - Upper Wear
    - Lower Wear
    - Dresses
- Validation
- Test

Proper folder organization reduces training errors and simplifies data management.

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10. Applications of Fashion Datasets

Fashion datasets are widely used in:

- Virtual Try-On Systems
- Smart Retail Platforms
- AI Shopping Assistants
- Clothing Recommendation Engines
- Fashion Search Systems
- Automated Product Tagging

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11. Research Findings

The study indicates that high-quality datasets are one of the most important components of AI-powered fashion applications.

DeepFashion2 provides comprehensive annotations for multiple computer vision tasks, while VITON-HD offers high-quality paired data for Virtual Try-On research.

Proper data cleaning, metadata generation, and annotation significantly improve model performance and reliability.

12. Learning Outcomes

During this research, the following concepts were studied:

- Fashion Datasets
- DeepFashion2
- VITON-HD
- Data Collection
- Data Cleaning
- Metadata Generation
- Image Annotation
- Dataset Organization

13. Conclusion

This research explored the role of fashion datasets in the development of Artificial Intelligence applications for the fashion industry. The study analysed DeepFashion2 and VITON-HD datasets and examined the processes involved in data collection, cleaning, annotation, and organization.

The findings demonstrate that well-prepared datasets form the foundation of successful Virtual Try-On systems and modern AI-powered fashion platforms. As AI technologies continue to evolve, the importance of high-quality data management will become even more significant.